

AoyuX Semi-annual Earthquake Forecasting Report for China Seismic Experimental Site (CSES)

version 1

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Semi-annual report for July 1, 2024 to December 31, 2024 (184 days)

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1. Earthquake Forecasts for the Next Six Months in the CSES

This report uses the earthquake catalog up to and including June 30, 2024, with a forecasting time window for the next half of the year (i.e., from July 1 to December 31, 2024, a total of 184 days). The magnitude type used in the catalog is local magnitude (M_L) released by the China Earthquake Networks Center (CENC). The simulations to obtain the earthquake forecasts are performed with the Epidemic Type Aftershock Sequence (ETAS) model with spatially varying background rate (Nandan et al., 2021a; 2021b, 2022; Li et al., 2024).

It is expected that, in the next six months, the level of seismicity within the CSES will not significantly change compared to the past six months (the first half of 2024). The probabilities of at least one earthquake of magnitude 5 or larger, and 6 or larger occurring in the region are 88% and 25%, respectively. Earthquakes of magnitude 5 or larger are likely to occur along the Longmenshan Fault, at the junction of the Longmenshan Fault and the Xianshuihe Fault, in the northern and southern parts of Yibin in Sichuan, in the area from Zhongdian to Yongsheng in Yunnan, in the area from Tengchong to Yongsheng in Yunnan, at the junction of the Zemuhe Fault and the Xiaojiang Fault, in the northern part of Dali in Yunnan and in Gengma in Yunnan, and other areas. One cannot rule out that an event of magnitude 6 or larger could occur in the Yongsheng area in Yunnan, which experienced two earthquakes of magnitude 4 or larger in the previous six months, southern Yibin in Sichuan, which experienced two earthquakes of magnitude 4 or larger in the previous six months, and at the junction of the Longmenshan Fault and the Xianshuihe Fault.

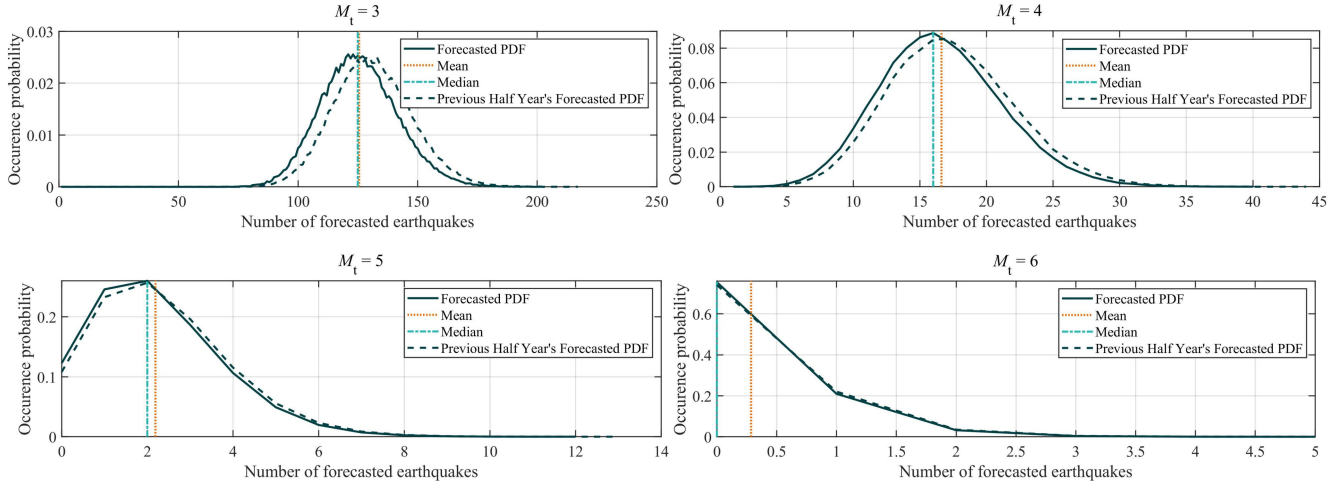


Figure 1. Forecasted probabilities of the numbers of earthquakes of different magnitude ranges ($M \geq M_t$) for the next half year.

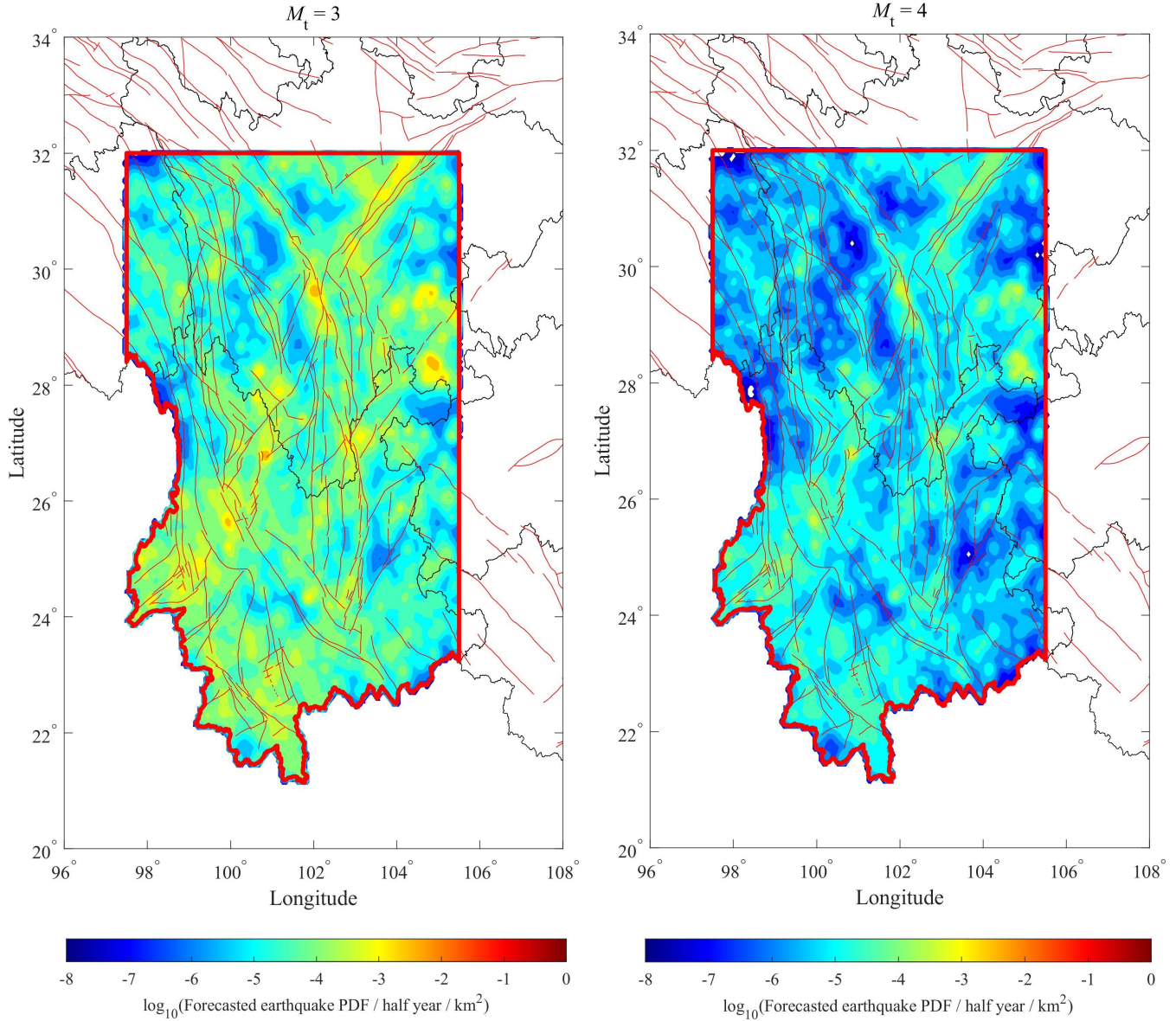


Figure 2. Probability density distribution (PDF) of the numbers of forecasted earthquakes ($M \geq M_t$) for the next half year per area (km^2) and per half year.

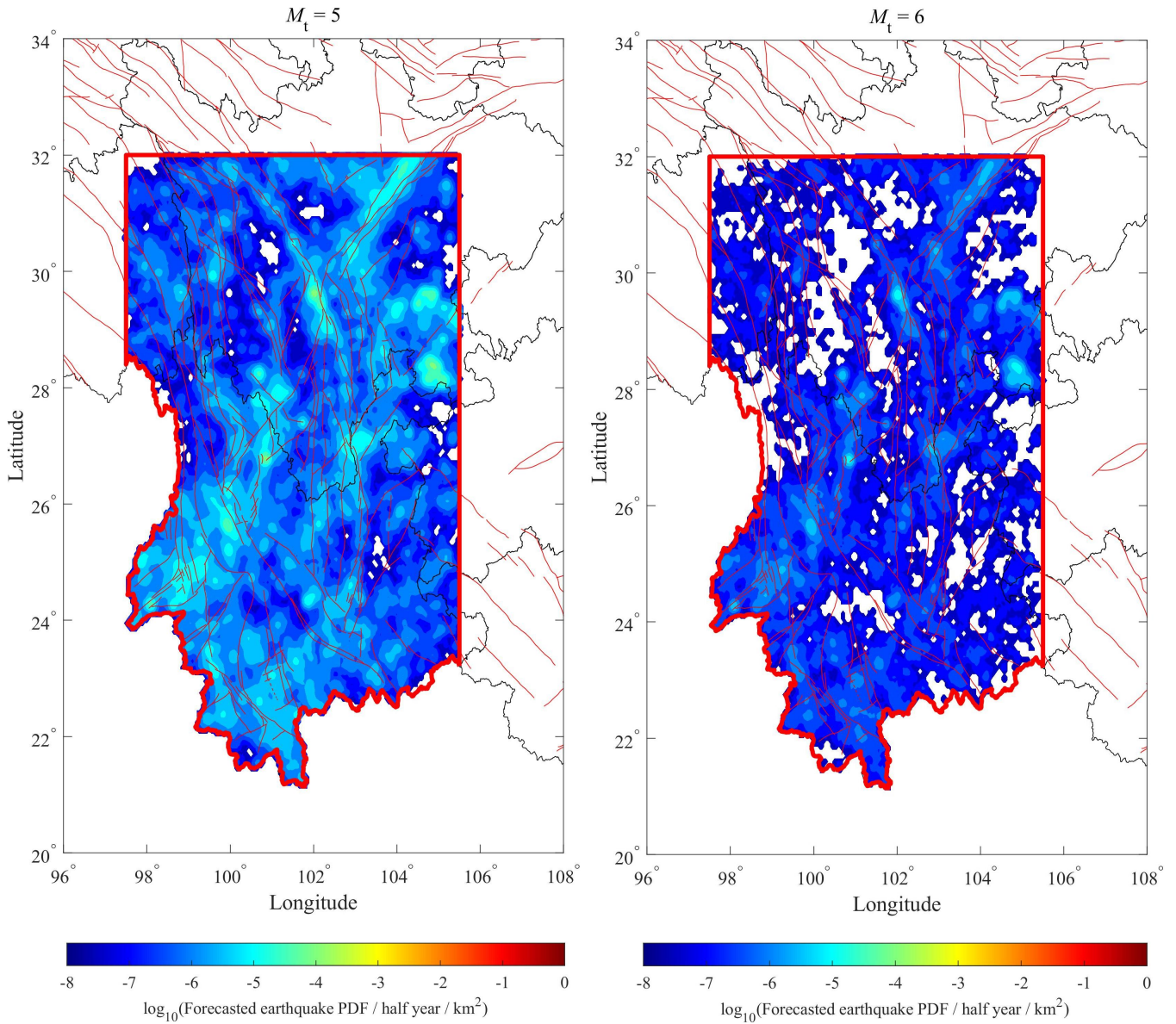


Figure 2. -continued.

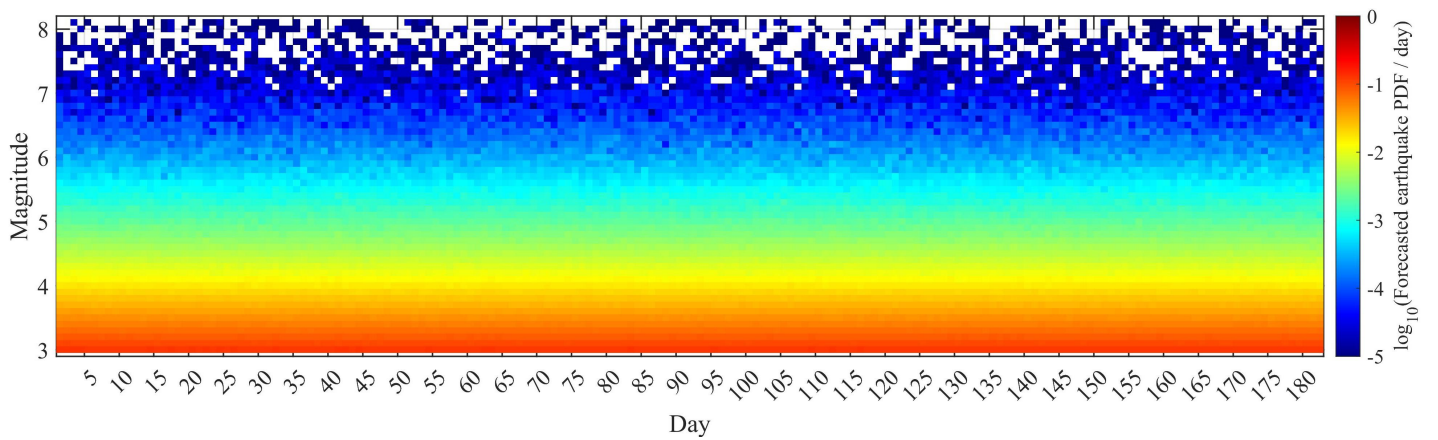


Figure 3. Daily probability density distribution (PDF) of the number of forecasted earthquakes per day over the whole CSES region for the next half year.

2. Previous Half Year's Forecast v.s. Realised Seismicity

In the previous half year (i.e., from January 1 to June 30, 2024, a total of 182 days), the region has experienced 97 earthquakes of $M \geq 3$, 4 of $M \geq 4$, and no earthquakes of $M \geq 5$ or above.

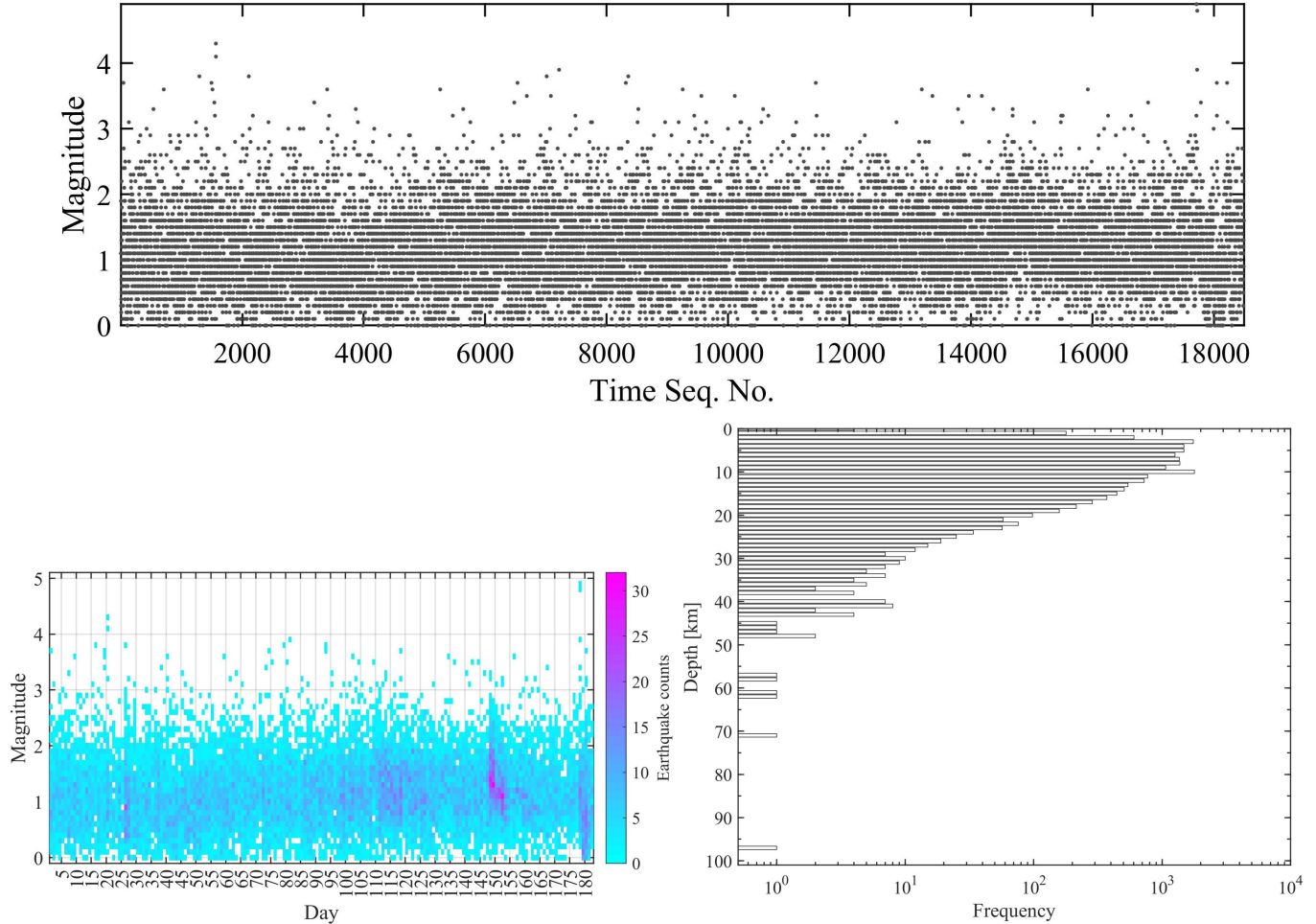


Figure 4. Seismicity in the CSES over the previous half year. (Top) Magnitude-time sequence plot; (Bottom left) Daily earthquake counts per 0.1 magnitude bin. (Bottom right) Depth-frequency distribution of earthquakes.

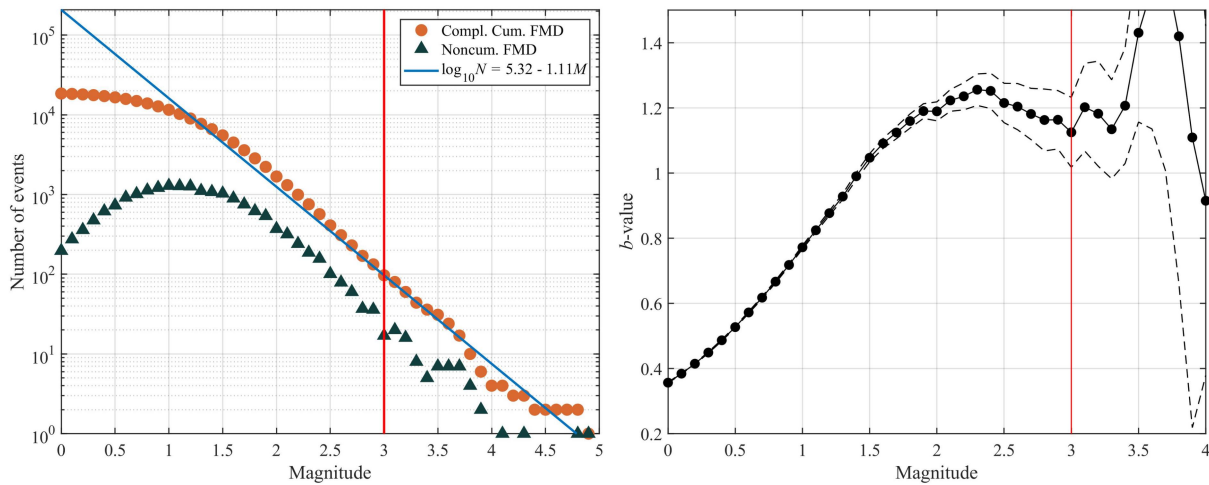


Figure 5. (left) Complementary cumulative and density frequency-magnitude distribution alongside the Gutenberg-ichter (GR) law fitted using earthquakes with magnitudes larger than the cut-off magnitude $M_{co} = 3.0$ (red line); (right) Variation of the b -value of the GR law with M_{co} , with the thin dashed line indicating the $\pm 1\sigma$ standard deviation boundaries. The seismicity analyzed is from the previous half year in the CSES.

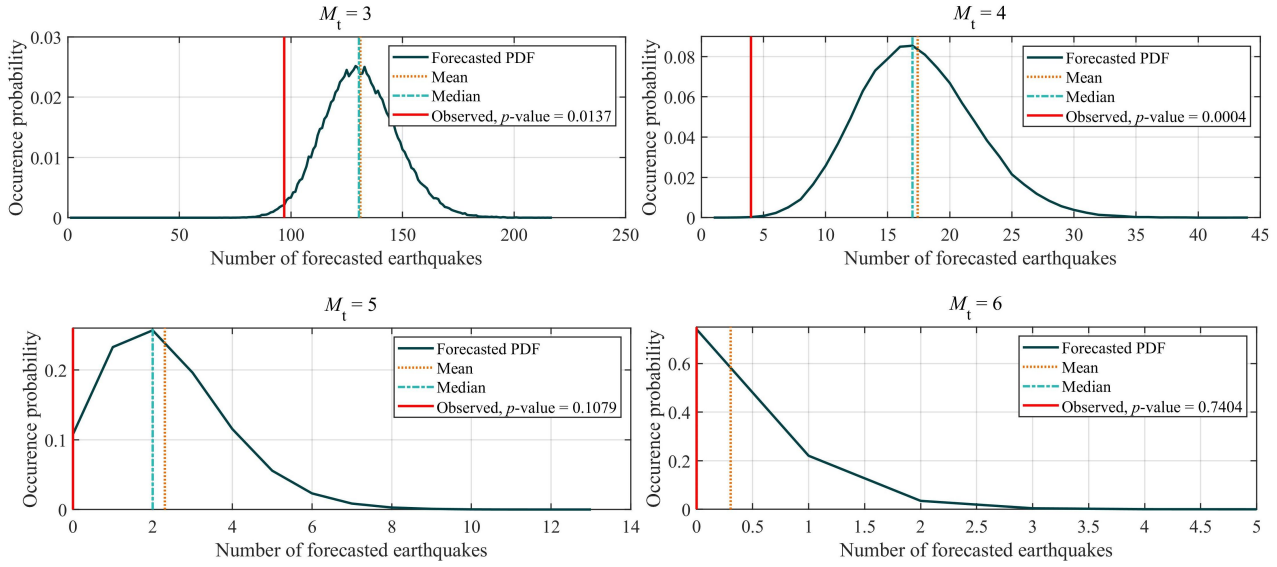
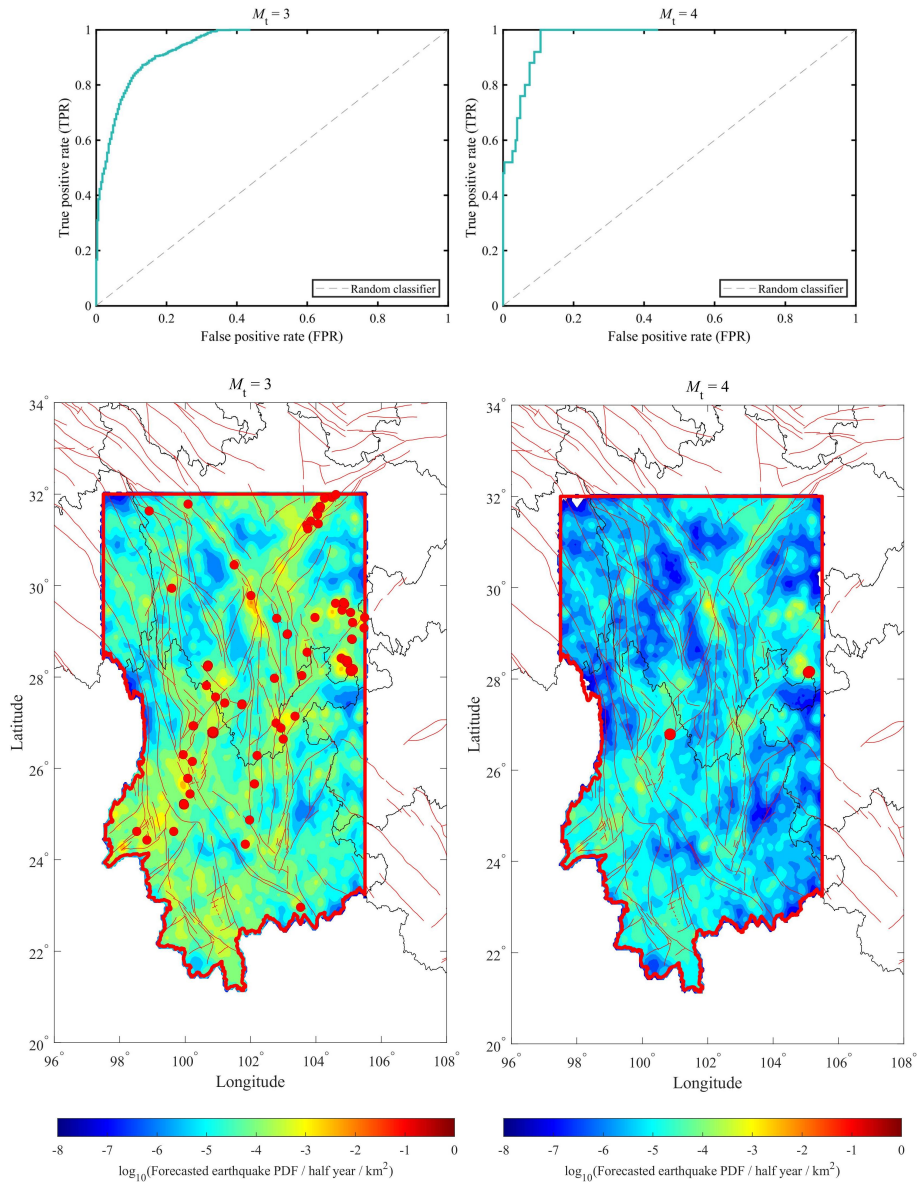


Figure 6. Number of observed earthquakes ($M \geq M_t$) in the CSES over the previous half year (red vertical line) and forecasted probabilities for earthquakes of different magnitude ranges ($M \geq M_t$) published the previous half year. The p -value is the cumulative distribution function (CDF) value for the number of observed earthquakes.



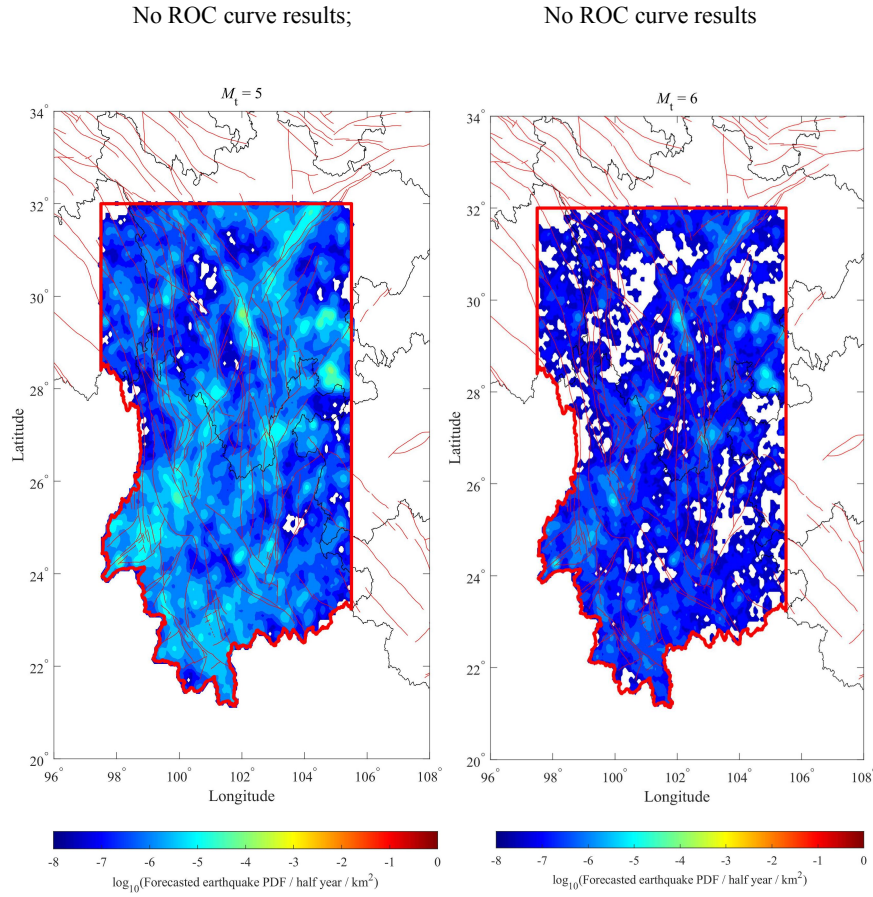


Figure 7. Top: Receiver Operating Characteristic (ROC) curve. The four following panels show the probability density distribution (PDF) of forecasted earthquakes ($M \geq M_1$) for the same period per area (km^2) and per half year, released by the last report as the background. Observed seismicity over the previous half year are shown as red solid circles. The ROC curve is constructed by iterating through all forecasted earthquake probability density values in descending order. For each probability density threshold, the true positive rate (TPR) and false positive rate (FPR) of the grid cells where actual earthquakes occurred are calculated. In this case, the ROC curve is built based on the probability density distribution of forecasted earthquakes ($M \geq M_1$) per area, as published in the last monthly forecast report. This process involves comparing the forecasted probability density with the actual earthquakes that occurred during that period to evaluate the performance of the prediction model.

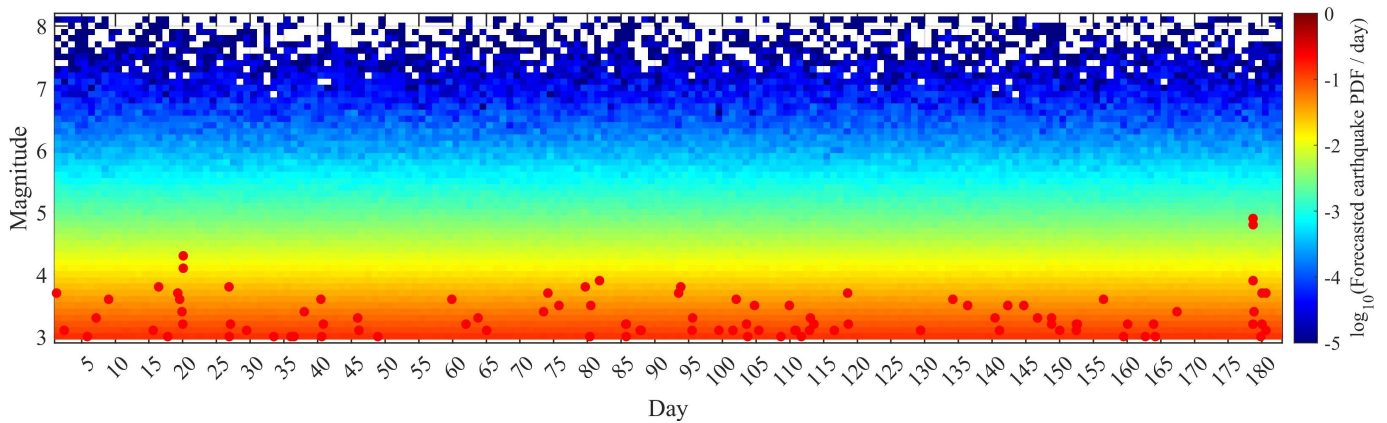


Figure 8. Observed seismicity (red solid circle) over the previous half year, with the daily probability density distribution (PDF) of the numbers of forecasted earthquakes for the same period, released by last report as the background.

3. Background Information

Since the inception of seismic records at the CSES (~ 780 BC), there have been 210 earthquakes of $M \geq 6$, 43 of $M \geq 7$, and 2 of $M \geq 8$. Since 1970, there have been 532 earthquakes of $M \geq 5$, 91 of $M \geq 6$, 14 of $M \geq 7$, and 1 of $M \geq 8$.

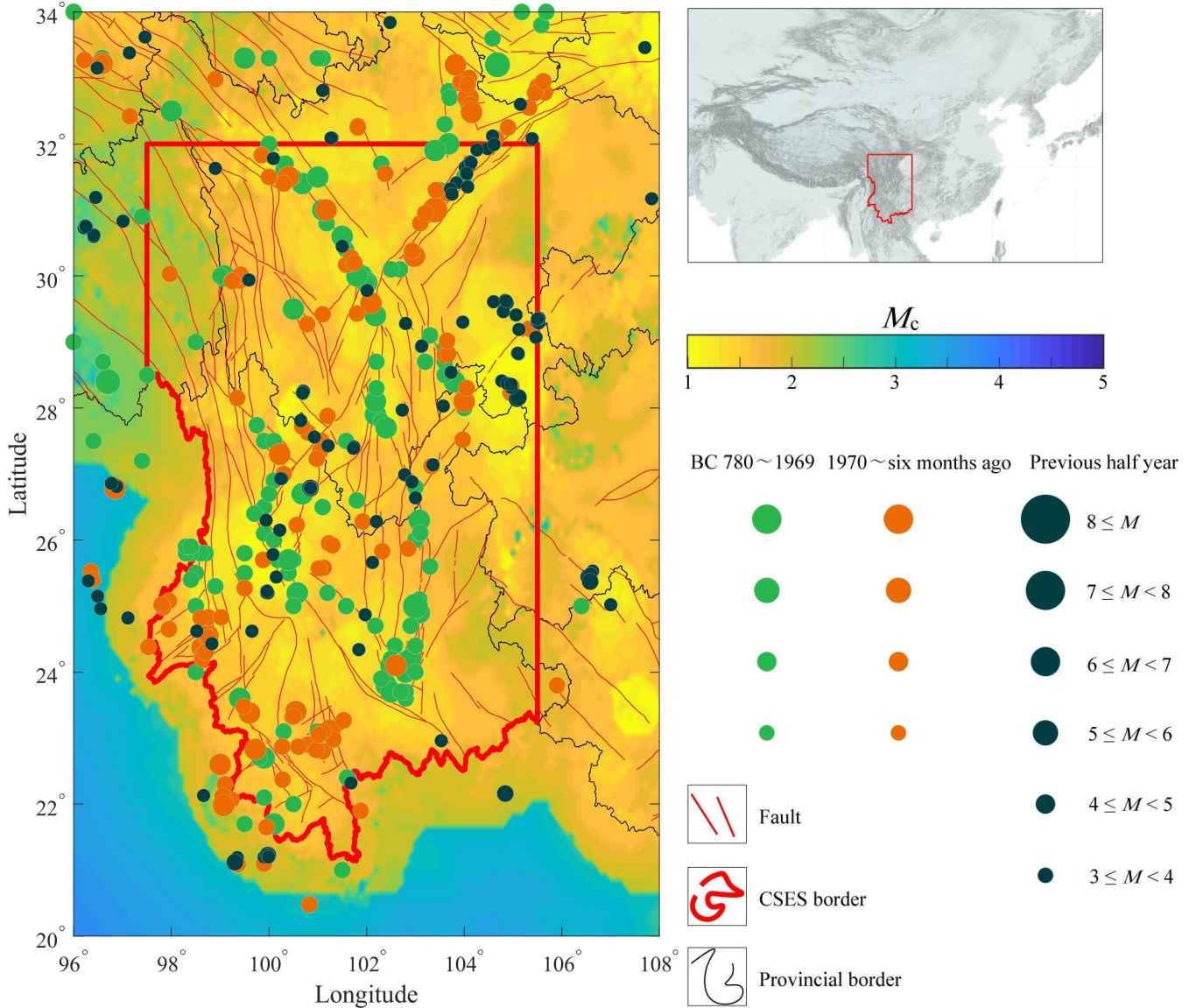


Figure 9. Spatial distribution of earthquakes and faults in the CSES located in the Sichuan-Yunnan region. The background map, showing the completeness magnitude (M_c), is adapted from Li et al. (2023).

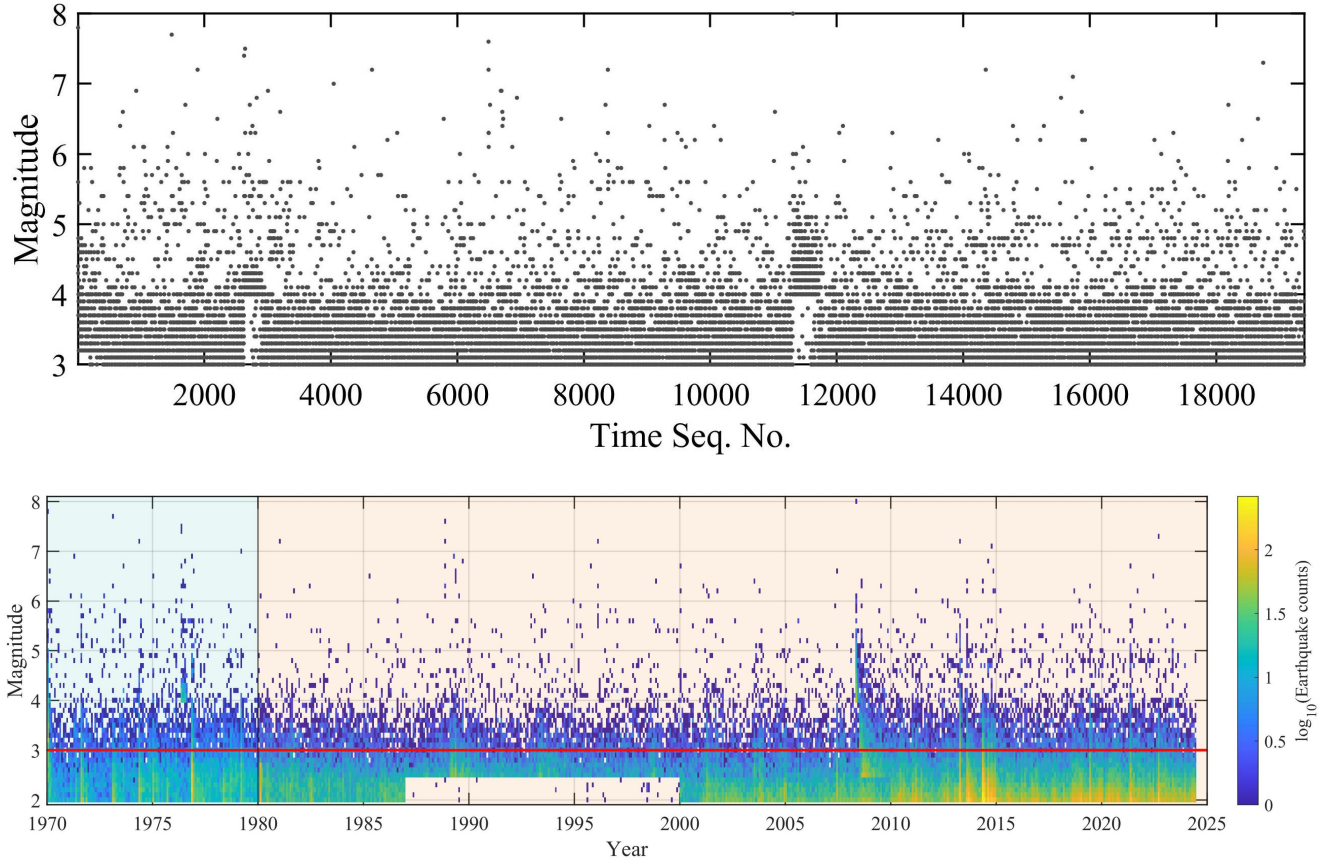


Figure 10. (Top) Magnitude-time sequence plot and (bottom) monthly earthquake counts per 0.1 magnitude bin in the CSES since 1970. The red line indicates the cut-off magnitude $M_{co} = 3.0$. The light red and light blue shaded regions represent the primary period and the auxiliary time band, respectively. The parameters M_{co} , primary period, and auxiliary period are all utilized for the ETAS model calibration and simulations to build the forecasts.

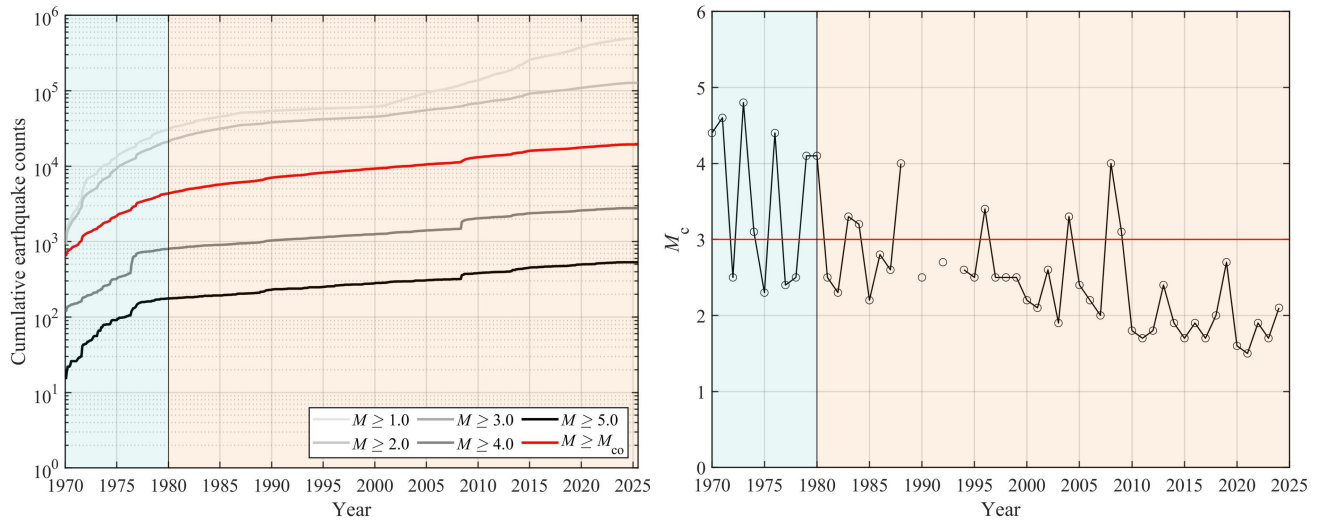


Figure 11. (left) Time series depicting cumulative earthquake counts for different cut-off magnitudes ($M_{co} = 1, 2, 3, 4$, and 5) in the CSES; (right) Evolution of the completeness magnitude (M_c) over time. The method to estimate M_c is described in Li et al. (2024).

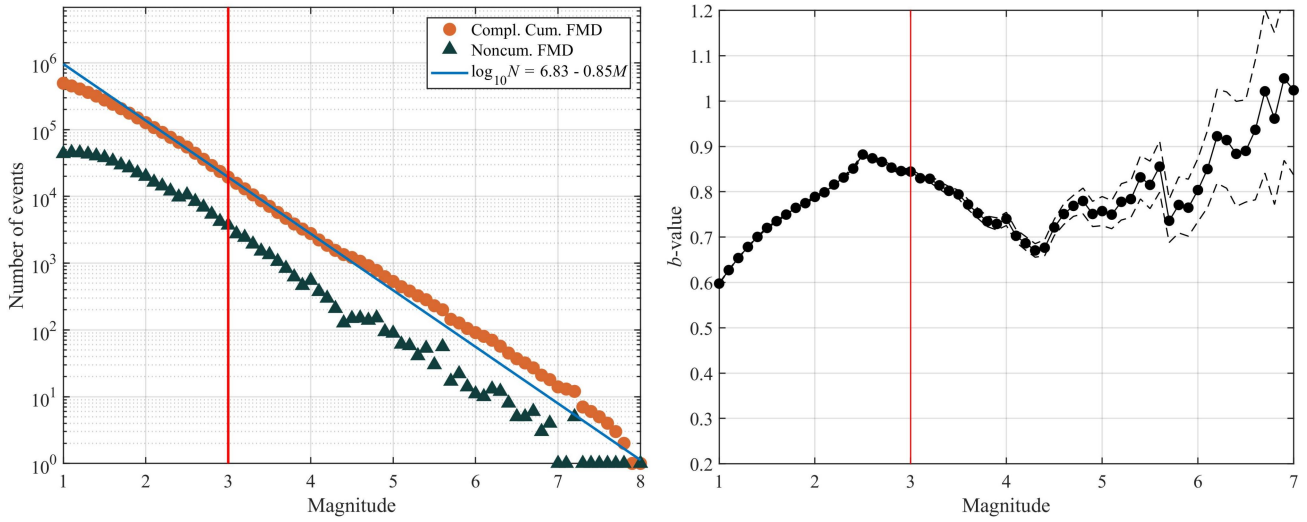


Figure 12. (left) Complementary cumulative and density frequency-magnitude distribution alongside the Gutenberg-Richter (GR) law fitted using earthquakes with magnitudes larger than $M_{co} = 3.0$ for the CSES catalogue since 1970; (right) Variation of the b -value of the GR law as a function of M_{co} , with the thin dashed line indicating the $\pm 1\sigma$ standard deviation. The methods to estimate b -value and to fit the GR law are described in Li et al. (2024).

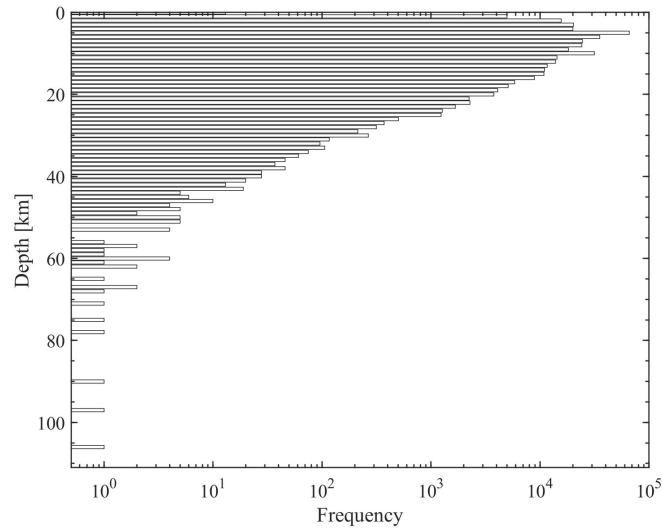


Figure 13. Depth-frequency distribution of earthquakes in CSES.

Acknowledgement

Since the first release of the AoyuX Earthquake Forecasting Series Report on June 4, 2024, we have received many constructive comments and suggestions from colleagues. These colleagues, in the order their feedback was received, include Prof. Jianchang Zhuang, Prof. Zhongliang Wu, Associate Prof. Yu Feng, Prof. Guixi Yi, Prof. Changsheng Jiang, and Prof. Yongxian Zhang. We express sincere gratitude to all of them! Most of these comments and suggestions have been reflected in the subsequently released series reports.

The earthquake catalogs in this report are sourced from the China Earthquake Networks Center (CENC) through the internal link provided by the Earthquake Cataloging System at the China Earthquake Administration, available at <http://10.5.160.18/console/index.action> (last accessed: December 18, 2023), with a Digital Object Identifier (DOI) of 10.11998/SeisDmc/SN. The Bayesian Magnitude Completeness (BMC) method used for calculating the spatial distribution of M_c was provided by Dr. Arnaud Mignan. The Epidemic Type Aftershock Sequence (ETAS) model code for simulating earthquake catalogs and calculating earthquake probabilities was derived from a code shared by Dr. Shyam Nandan which is used in (Nandan et al., 2021a;b; 2022). Fault data were obtained from the GMT Chinese Community at <https://docs.gmt-china.org/latest/dataset-CN/CN-faults/> (last accessed: May 31, 2024). This work is partially supported by the National Natural Science Foundation of China (Grant no. U2039202 and U2039207), the Guangdong Basic and Applied Basic Research Foundation (Grant No. 2024A1515011568), the Special Fund of the China Seismic Experimental Site 2022 (DQJB22Z01-10), the Shenzhen Science and Technology Innovation Commission (Grant no. GJHZ20210705141805017), and the Center for Computational Science and Engineering at Southern University of Science and Technology.

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Attachment: About AoyuX

AoyuX is a developing online research platform dedicated to advancing earthquake predictability in China. It is produced by the Institute of Risk Analysis, Prediction, and Management (RisksX) at the Southern University of Science and Technology (SUSTech), in collaboration with multiple research institutions. In Chinese mythology, "Aoyu" is a legendary creature with a dragon's head and a fish's body, believed by ancient Chinese to cause earthquakes. The "X" represents the unknown and infinite in Western culture, symbolizing goals and hopes.

The AoyuX platform aims to use cutting-edge statistical seismology tools for forward earthquake forecasting research. It will also serve as an online research platform for forward earthquake forecasting competitions with other models. In the future, the AoyuX platform will integrate geophysical observation data, artificial intelligence (AI), and other technologies to conduct various types of earthquake forecasting research, including statistical, numerical, and physical methods. AoyuX is dedicated to help building a modern earthquake forecasting evaluation system in China, encompassing mid-term, short-term and imminent forecasts. It will develop a new visualization platform for earthquake predictability research, facilitating communication between experts and the public, and serving the combined efforts of experts and the public in tracking and assessing the imminent occurrence of major earthquakes, as well as in mid-term and short-term earthquake forecasting.

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